RAJALAKSHMI ENGINEERING COLLEGE

Lathikka D 251001133-2025-IT

L2

Started on	Friday, 31 July 2026, 4:21 PM
State	Finished
Completed on	Friday, 31 July 2026, 4:43 PM
Time taken	22 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int c=0;
    int count=0;

    if(scanf("%d",&n) !=1){
        return 0;
    }
    count++;
    for(int i=n/2;;){
        count++;
        if(!(i<n)){
            break;
        }
        int j=1;
        for(;;){
            count++;
            if(!(j<n)){
                break;
            }
            int k=1;
            for(;;){
                count++;
                if(!(k<n)){
                    break;
                }
                c++;
                count++;

                k=k*2;
            }
            j=2*j;
        }
        i++;
    }
    printf("%d\n",count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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